

Math Module

Python math module is defined as the most famous mathematical functions, which includes trigonometric functions, representation functions, logarithmic functions, etc.

1. math.pow()

This method returns the power of the x corresponding to the value of y.

If value of x is negative or y is not integer value than it raises a ValueError.

```
import math
number = math.pow(10,2)
print("The power of number:",number)
```

output:

The power of number: 100.0

2. math.floor()

This method C It returns the less than or equal value to x.

```
import math
number = math.floor(10.25201)
print("The floor value is:",number)
```

output:

The floor value is: 10

3. math.ceil()

This method returns the ceil value of the x. It returns the greater than or equal value to x.

```
import math
```

```
number = math.ceil(10.25201)
print("The ceil value is:", number)
```

output:

The ceil value is: 11

4. math.factorial()

This method returns the factorial of the given number x. If x is not integral, it raises a ValueError.

```
import math
number = math.factorial(7)
print("The factorial of number:", number)
```

output:

The factorial of number: 5040

5. math.modf()

This method returns the fractional and integer parts of x. It carries the sign of x is float.

```
import math
number = math.modf(44.5)
print("The modf of number:", number)
```

output:

The modf of number: (0.5, 44.0)

6. math.sqrt()

This method returns the square root of the given number

```
import math
x = math.sqrt(64)
print(x)
```

output:

8

7. math.pi constant, returns the value of PI

```
import math
x = math.pi
print(x)
```

output:

3.141592653589793

Random Module

Random module is an in-built module of Python which is used to generate random numbers. These are pseudo-random numbers means these are not truly random. This module can be used to perform random actions such as generating random numbers, print random a value for a list or string, etc.

1. random.choice()

Printing a random value from a list

```
import random
# prints a random value from the list
list1 = [1, 2, 3, 4, 5, 6]
print(random.choice(list1))
```

output:

2

2. random.randint()

random.randint() method is used to generate random integers between the given range.

```
import random
# Generates a random number between a given positive range
r1 = random.randint(5, 15)
print("Random number between 5 and 15 is %d" % (r1))

# Generates a random number between two given negative range
r2 = random.randint(-10, -2)
print("Random number between -10 and -2 is %d" % (r2))
```

output:

Random number between 5 and 15 is 13

Random number between -10 and -2 is -8

3. random.random()

random.random() method is used to generate random floats between 0.0 to 1.0

```
from random import random
# Prints random item
print(random())
```

output:

0.9876107530027021

4. random.shuffle()

random.shuffle() method is used to shuffle a sequence (list). Shuffling means changing the position of the elements of the sequence.

```
import random
# declare a list
sample_list = [1, 2, 3, 4, 5]
print("Original list : ")
print(sample_list)
# first shuffle
random.shuffle(sample_list)
print("\nAfter the first shuffle : ")
print(sample_list)

# second shuffle
random.shuffle(sample_list)
print("\nAfter the second shuffle : ")
print(sample_list)
```

output:

Original list :
[1, 2, 3, 4, 5]

After the first shuffle :
[3, 4, 5, 2, 1]

After the second shuffle :
[4, 1, 2, 5, 3]

Statistics module

Python statistics module provides the functions to mathematical statistics of numeric data. There are some popular statistical functions defined in this module.

1. statistics.mean()

The statistics.mean() function is used to calculate the arithmetic mean of the numbers in the list.

```
import statistics
# list of positive integer numbers
datasets = [5, 2, 7, 4, 2, 6, 8]
x = statistics.mean(datasets)
# Printing the mean
print("Mean is :", x)
```

output:

Mean is : 4.857142857142857

2. statistics.median()

The median() function is used to return the middle value of the numeric data in the list.

```
import statistics
datasets = [4, -5, 6, 6, 9, 4, 5, -2]
# Printing median of the
# random data-set
print("Median of data-set is : % s " %
      (statistics.median(datasets)))
```

output:

Median of data-set is : 4.5

3. statistics.mode()

The mode() function returns the most common data that occurs in the list.

```
import statistics
# declaring a simple data-set consisting of real valued
positive integers.
dataset =[2, 4, 7, 7, 2, 2, 3, 6, 6, 8]
# Printing out the mode of given data-set
print("Calculated Mode % s" % (statistics.mode(dataset)))
```

output:

Calculated Mode 2

4. statistics.stdev()

The stdev() function is used to calculate the standard deviation on a given sample which is available in the form of the list.

```
import statistics
# creating a simple data - set
sample = [7, 8, 9, 10, 11]
# Prints standard deviation
print("Standard Deviation of sample is % s
"%(statistics.stdev(sample)))
```

5. median_low()

The median_low function is used to return the low median of numeric data in the list.

```
import statistics
# simple list of a set of integers
set1 = [4, 6, 2, 5, 7, 7]
# Note: low median will always be a member of the data-set.
# Print low median of the data-set
print("Low median of data-set is % s " %
(statistics.median_low(set1)))
```

output:

Low median of data-set is 5

6. median_high()

The median_high function is used to return the high median of numeric data in the list.

```
import statistics
# list of set of the integers
dataset = [2, 1, 7, 6, 1, 9]
print("High median of data-set is %s " %
      (statistics.median_high(dataset)))
```

output:

High median of data-set is 6



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